



Shay locomotive No. 25 with a carriage rake.

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Alishan Forest Railway

Taiwan's 762 mm forest and mountain railway from Chiayi to Alishan, with active sunrise and forest-park branches.

Lines

The scheduled Chiayi–Alishan main line, forest-park branches and the closed Mianyue Line.

Alishan Forest Railway

A 762 mm mountain railway from Chiayi to Alishan, with spiral, Ω -shaped and Z-shaped climbing works, Shay locomotives and active forest-park branches.

Stations

A source-separated inventory of the Alishan main line and forest-park branch points.

Alishan Forest Railway stations

The current 16-stop main-line timetable and four named forest-park branch points, with older heritage and mileage inventories kept visible as separate records.

Rolling stock

The Alishan Forest Railway's Shay steam locomotives, diesel fleet, and incoming next-generation train.

Alishan Forest Railway rolling stock

Four Shay geared steam locomotives still turn under their own power, according to one operator page — or just one, according to another. Both are published here, unresolved, alongside a next-generation train now arriving and a Wikipedia citation that doesn't check out against the source it names.

Operations

Alishan Forest Railway fares, booking, and a decade of ridership data.

Alishan Forest Railway fares and ridership

A flat NT\$600 to ride the full mountain line, worked out from a published per-kilometre rate — and a decade of monthly ridership data this project extracted directly from the operator's own PDF, showing the branch lines carrying five to ten times the main line's traffic every single year.

The Alishan Forest Railway is a working heritage railway, not a scenic line invented after the forest industry ended. The operator dates its opening to 1912 and describes a main line of about 72 km with a network that once grew to 21 branches.^[1] Its current public timetable runs from Chiayi to Alishan, while the forest-park branches continue the railway beyond the main-line terminus toward Shenmu, Zhaoping and Zhushan.^[3] ^[4]

The railway is unusually legible as a system because the operator publishes route technology and station information together. Its gauge is 762 mm, and the line climbs from Chiayi at 30 m above sea level to Alishan at 2,216 m before the Zhushan branch reaches 2,451 m.^[1] ^[2] The line is non-electrified and its engineering vocabulary is a response to tight mountain land, not a decorative theme: the official heritage record identifies spiral, horseshoe, Ω/U-turn and Z-shaped alignments alongside the special Shay geared locomotive.^[5]

The line should be read in two layers. The 16-stop main line is the current intercity-style forest-railway service between Chiayi and Alishan, with a timetable effective from 10 January 2025.^[3] The branch timetable is a separate operating pattern inside the Alishan National Forest Recreation Area: Shenmu and Zhaoping have frequent short workings, while the Zhushan service is an observation train whose departure is announced the day before because it is tied to sunrise.^[4] Duigaoyue is a named platform and passing location on the Zhushan alignment in the operator's station overview, but the branch timetable does not establish it as a separately scheduled passenger stop.^{[2] [4]}

This distinction also explains why station counts vary between pages. The current operator FAQ enumerates 16 main-line stations and four named forest-park points, while the heritage potential-site description uses an older landscape boundary and a different list of 16 named stations.^{[8] [5]} The guide keeps the inventories side by side in the station data rather than treating a heritage boundary as a current timetable. The older operation-mileage page adds another source view: its matrix omits Luma Chan and the branch points, even though the current operator overview and timetable include Luma Chan and the park services.^{[9] [2] [3]}

The railway's most famous engineering is concentrated in the climb. At Duli Shan, the track circles the mountain in a three-turn spiral; at Shuisheliao, the Ω -shaped alignment turns the train through 180 degrees; and beyond Pingzhena the railway uses Z-shaped reversals because the mountain shelf is too narrow for a normal curve.^{[5] [2]} The operator describes the Shay design in the same practical terms: vertical cylinders, bevel gears and flexible drive connections let a short locomotive negotiate tight curves, tunnels and steep grades while relying on adhesion rather than a rack rail.^{[10] [5]}

The rolling stock records are a roster of families rather than a set of thin vehicle biographies. The operator identifies early 13-ton Shay locomotives, eight 18-ton two-cylinder locomotives bought between 1910 and 1913, and twelve 28-ton three-cylinder locomotives bought between 1912 and 1917.^[10] Its current steam-locomotive page says Nos. 25 and 31 were repaired and returned to running after conversion from coal to diesel firing, with the repairs completed in 2006 and 2004 respectively.^[11] The operator's FAQ separately places No. 26 in static display and says the surviving display and occasional-running locomotives can be seen at the Chiayi and Fenqihu facilities or on the mountain.^[8] Diesel power is the everyday roster: the operator's table lists active DL25–30, DL31–34, DL38, DL39–43 and DL45–51 groups, alongside retired early Mitsubishi and O&K groups.^[12]

Typhoon damage is part of the railway's operating history. Typhoon Morakot damaged the system in 2009, and Typhoon Dujuan caused two collapses totalling 55 m in the old No. 42 tunnel between Shizilu and Pingzhena in 2015.^{[6] [7]} The replacement 1,106 m tunnel began construction on 8 January 2021 and was completed on 29 December 2023, according to the railway's project record.^[7] That is an infrastructure milestone; the scheduled passenger reopening was separately announced for 6 July 2024 after Railway Bureau inspection, with four daily main-line services in the opening timetable.^{[6] [3]} The distinction matters because "completed" and "open to passengers" are not the same date.

The line is a Taiwanese World Heritage potential site, not an inscribed UNESCO World Heritage property. The Ministry of Culture's heritage page presents the railway and its surrounding cultural landscape as a potential site, describes its 762 mm non-electrified mountain railway, and gives a 71.34 km potential-site scope across Chiayi and Nantou.^[5]

The same record compares the route's industrial settlements, forest ecology and mountain engineering with the values usually used to assess railway heritage.^[5] In 2019, the government recreation history records the railway's registration as an important cultural landscape, which is a domestic heritage status distinct from UNESCO inscription.^[13]

This system page therefore links to a sourced line record and one station-inventory page. It does not create twenty short articles whose only content would be a name and elevation: the official sources provide a strong route-level history and a useful station table, but not a uniform 300-word engineering or accessibility record for every stop.^[2]^[8]

The rolling-stock record `[/rail/alishan/rolling-stock/fleet-roster/]` goes deeper on the Shay fleet than this page does, including a conflict between the operator's own pages over how many locomotives can still run, and a Wikipedia citation that doesn't check out against the source it names. The fares and ridership record `[/rail/alishan/operations/fares-and-ridership/]` covers the distance-based fare formula and a full decade of monthly ridership this project extracted directly from an operator PDF an earlier research pass couldn't open.

References

13 of 13 sources on this page are primary — published by the operator, the builder or government.

- Railway technology** `[https://afrch.forest.gov.tw/0000524]`
鐵道技術
PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處) accessed 2026-08-25
The operator's technical history gives the 1912 opening, approximate main-line length, 762 mm gauge, earlier branch count and the spiral, U-shaped and Z-shaped mountain methods.
- Alishan Forest Railway station overview** `[https://afrch.forest.gov.tw/]`
阿里山百年鐵道
PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處) accessed 2026-08-25
The operator's current station overview gives the published station names, chainages and elevations and describes the forest-park branch points.
- Main-line timetable** `[https://afrch.forest.gov.tw/0000115]`
林鐵本線
PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處) accessed 2026-08-25
The timetable effective 10 January 2025 gives the currently published main-line service sequence and daily train times.
- Branch-line timetable and closure notes** `[https://afrch.forest.gov.tw/0000300]`
林鐵支線
PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處)

) accessed 2026-08-25

The current branch page gives the Shenmu, Zhaoping and Zhushan service records, the Mian Yue closure history and the 6 July 2024 timetable boundary.

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- 5 **Alishan Forestry and Railway Cultural Landscape** [https://twh.boch.gov.tw/taiwan/intro.aspx?id=7&lang=zh_tw]

阿里山林業暨鐵道文化景觀

PRIMARY Bureau of Cultural Heritage, Ministry of Culture (文化部文化資產局)

accessed 2026-08-25

The official Taiwan World Heritage potential-site page describes the route's heritage status, engineering methods, 71.34 km potential-site boundary and 32,900-hectare landscape.

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- 6 **Alishan Forest Railway full-line opening on 6 July 2024** [<https://www.forest.gov.tw/news/0073945>]

阿里山林鐵113年7月6日全線通車

PRIMARY Forestry and Nature Conservation Agency, Ministry of Agriculture (

農業部林業及自然保育署) accessed 2026-08-25

The agency's full announcement records the Morakot and Dujuan typhoon damage, the 15-year interruption, the 6 July 2024 passenger opening, four daily main-line trains and the NT\$600 Chiayi–Alishan fare.

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- 7 **No. 42 tunnel completion ceremony** [<https://afrch.forest.gov.tw/AllNews/0073708>]

林鐵42號隧道竣工典禮 祈願一路順遂

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處

) accessed 2026-08-25

The official project notice gives the 55 m collapse, the new 1,106 m tunnel, construction start on 8 January 2021 and the engineering and environmental constraints.

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- 8 **Frequently asked questions** [<https://afrch.forest.gov.tw/faqs?iPage=2>]

常見問答

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處

) accessed 2026-08-25

The operator's current FAQ lists the 16 main-line stations and the named forest-park points and describes the Shay display and running locations.

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- 9 **Operation mileage** [<https://afrch.forest.gov.tw/0000120>]

營業里程

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處

) accessed 2026-08-25

The operator's mileage matrix, updated 31 March 2016, is retained as a dated historical measurement view.

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- 10 **Shay steam locomotive history** [<https://afrch.forest.gov.tw/0000107>]

夏依式蒸汽機車

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處

) accessed 2026-08-25

Operator history for the LIMA Shay classes and their use during railway construction.

11 Introduction of steam locomotive [<https://afrch.forest.gov.tw/EN/0000108>]

Introduction of Steam Locomotive

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處) accessed 2026-08-25

Official English mirror for the No. 25 and No. 31 repair and diesel-firing conversion record; the Chinese page timed out during this check.

12 Introduction of diesel locomotive [<https://afrch.forest.gov.tw/0000106>]

柴油機車之介紹

PRIMARY Alishan Forest Railway and Cultural Heritage Office (阿里山林業鐵路及文化資產管理處) accessed 2026-08-25

Operator table of diesel locomotive groups, manufacturers, build years and current or retired status.

13 Alishan Forest Railway [https://recreation.forest.gov.tw/Forestry/FR?typ=11&typ_id=0900041]

阿里山林業鐵路

PRIMARY Forestry and Nature Conservation Agency recreation site (林業及自然保育署台灣山林悠遊網) accessed 2026-08-25

Government chronology for the 2019 important cultural landscape registration and the 2023/2024 reopening milestones.

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Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, TYMC, and TMRT. Government open data. Retrieved 2026-08-24; source last updated 2016-11-23. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.